INCH-POUND

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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY M45A1 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M45A1 Delay Cartridge Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

MIL-I-14804 Initiator, Cartridge Actuated, Delay, M4A1, M12A1, M45A1, and M49A1 Remanufactured Metal Parts Assembly

MIL-I-46497 Initiator, Cartridge Actuated, Delay, M4A1, M12, M45A1, and M49 Metal Parts Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Unless otherwise indicated, copies of military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

8596110	Cartridge, Delay, M114 Assembly
8596101	Initiator, Cartridge Actuated, Delay, M45A1 Assembly
8796826	Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1 or M49A1
9216824	Box, Fiberboard (Outer) Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1 or M49A1

U.S. Air Force

TPO 00-845-1058	Initiator, Cartridge Actuated, M26

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document,

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however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Conformance to documents. The M45A1 initiator covered by this specification shall be manufactured in accordance with DL8596101.

3.3 Components.

3.3.1 Primary components. For the purpose of this specification, the M114 delay cartridge (Drawing 8596110) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot.

3.3.2 Component parts. All component parts shall meet the applicable requirements of MIL-I-46497 or MIL-I-14804 prior to assembly.

3.4 Initiator pin security. Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanment deformation of the safety pin (see 4.3.1).

3.5 Breakaway torque. The chamber and cap of each assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.3.2).

3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches (see 4.3.3).

3.7 Radiographic examination. Each initiator assembly shall be radiographically examined for defects (see 4.3.4).

3.8 Functional.

3.8.1 Ballistics. Initiators loaded with cartridges and temperature conditioned shall perform within the pressure values specified in table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, assembly number MS28741-4, and a pressure chamber with a volume of 1 cubic inch.

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3.8.2 Misfires. There shall be no misfires.

3.8.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.4.2).

TABLE I. Pressure values.

Temperature (°F)	Pressure (psig)		Delay time, seconds (see 6.4.1)
	Minimum	Maximum	
-60 to -70	1300	2200	3.5 + 0.7, - 0.8
65 to 75	1300	2200	3.0 ± 0.3
195 to 205	1300	2500	2.700 ± 0.006

3.9 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation, and identification of lots shall be in accordance with MIL-STD-105.

4.2.1.2 Size of lot. The production lot size shall be as specified in the contract (see 6.2).

4.2.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.2.2.1 Cartridge, Delay, M114, Drawing 8596110, covering a detail of Drawing 8596101.

Characteristics	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	Marking missing, incorrect, or illegible	Visual
Major		
101	Foreign matter present	Visual
Minor		
201	Case scratched, wrinkled, bulged, or dented	Visual
202	Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing 8596101, less Drawings 8596110, 8593313, 8593114, and MIL-C-5501/11.

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Critical		
1	Mixed initiators	Visual
2	Initiator pin not in cocked position	Visual
3	Safety pin missing or improperly engaged	Visual
Major		
101	O-ring missing or not lubricated	Visual
102	Thread damaged	Visual
103	Burr	Visual
104	Foreign matter	Visual
Minor		
None defined		

4.2.2.3 Assembly, Drawing 8596101, less Drawing 8593314 and MIL-C-5501/11.

Critical		
1	Cartridge loose	Manual/Visual
2	Cartridge retainer loose	Manual/Visual
3	Metering groove of cartridge retainer missing	Visual
4	Elongated hole of cartridge retainer missing	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

4.2.2.4 Assembly, Drawing 8596101, less MIL-C-5501/11.

Critical		
1	Location of cartridge retainer	X-ray
2	Gap between cartridge retainer and chamber	X-ray
3	Gap between firing pin housing and cap	X-ray
4	Foreign matter in gas port	Visual
Major		
101	Length from cap boss face to gas port face, max	Gage
102	Protrusion of firing pin, max	Gage
103	Set screw missing, loose, or above flush	Manual/Visual
Minor		
None defined		

4.2.2.5 Assembly, Drawing 8596101.

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Critical		
1	Base of firing pin housing not in complete contact with face of cartridge	Manual/X-ray
2	Any part missing, except as otherwise classified	Visual/X-ray
3	Improper assembly	Visual/X-ray
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Height of cap, max Gage	
102	Initiator pin not free to rotate	Manual
103	Safety pin loose or deformed	Manual/Visual
104	Setscrew not covered with barrier material	Visual
Minor		
201	Total length, max Measure	
202	Location of marking incorrect	Visual
203	Shipping cap missing Visual	
204	Protective finish inadequate or defective	Visual

4.2.2.6 Fiber Container, Drawing 8796826.

Critical
None defined

Major
None defined

Minor		
201	Contents loose Manual	
202	Box damaged Visual	
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.2.2.7 Box, Ammunition, Packing, Drawing 8796827.

Critical
None defined

Major
None defined

Minor		
201	Contents loose Manual	
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual

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203 Contents of box exposed Visual
204 Box damaged Visual
205 Marking missing, incorrect, or illegible Visual
206 Number of units incorrect Visual

4.2.2.8 Box, Fiberboard (Outer), Alternative, Ammunition, Packing, Drawing 9216824.

Critical
None defined

Major
None defined

Minor
201 Contents loose Manual/visual
202 Box broken or damaged Visual
203 Number of units contained incorrect Visual
204 Marking missing, incorrect, or illegible Visual
205 Tape incomplete Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any unit to comply with the requirement of 3.4 shall be cause for rejection of the subassembly.

4.2.3.2 Breakaway torque. Each chamber and cap assembly shall be subjected to the breakaway torque test. Failure of any assembly to comply with 3.5 shall be cause for rejection of the assembly.

4.2.3.3 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.3. Failure of any initiator to comply with 3.6 shall be cause for rejection of the initiator.

4.2.3.4 Functional tests. The following tests shall be performed by the Government.

4.2.3.4.1 Ballistics. Twenty-one units from each lot shall be tested to determine compliance with 3.8. The units shall be temperature conditioned as provided in table II. Failure of any unit to comply with 3.8 shall be cause for rejection of the lot.

TABLE II. Conditioning temperature.

Temperature (°F)	Sample Size
-60 to -70	7
65 to 75	7

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195 to 205	7

4.2.3.4.2 Retest. There shall be no retests.

4.2.3.4.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.3 Test methods and procedures.

4.3.1 Initiator pin security. A pull force shall be applied to the initiator pin in the longitudinal direction in conjunction with the applicable fixture which registers the amount of pull.

4.3.2 Breakaway torque. The torque wrench set as specified in 3.5 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in an air vise or holding fixture. The torque shall be applied to the cap.

4.3.3 Residual magnetism. The initiator shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the initiator 5 inches from the end, and at the same height as the compass, in the north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.4 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with MIL-STD-453, with the exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The initiators shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Cartridge, delay M114	P/N 8596110
b. Firing pin	P/N 8595407
c. Chamber	P/N 8593314

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d. Firing pin housing	P/N 8595408
e. Cartridge retainer	P/N 8593313
f. CAP	P/N 8595404
g. 3-Ball	MS19060-3
h. Spring, initiator	P/N 8595406
i. Pin, initiator	P/N 8595403
j. O-ring	MS28775-214
k. Pin, safety	P/N 8595405

4.3.5 Functional.

4.3.5.1 Ballistics. The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned. Required pressure readings shall be taken as specified in 3.8.1.

4.3.5.2 Temperature conditioning. Each assembled initiator shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -65°F, all condensation shall be removed from the exterior surfaces before returning the initiator to the conditioning chamber.

4.4 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8796826 and 8796827. Level B shall be in accordance with Drawings 8796826 and 9216824.

5.1.1 Lot numbers. Lot numbers shall be in accordance with MIL-STD-1168.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-845-1058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification

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b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. Production test activity and lot size (see 4.1 and 4.2.1)

d. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.3.4)

e. Level of packing and marking (see 5.1)

f. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of ``Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Hazard notice. The initiator described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 Definitions.

6.4.1 Delay time. Delay time is the time from primer impact to the start of pressure rise.

6.4.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

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Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E57)

Review activity:
Air Force - 70

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